CSA3710 SOFTWARE TESTING
EXPERIMENTS

Name: Kaviya V **Register No:** 192421022

QUESTIONS:

2. To write a case study to identify & analyze the requirements of ARMS Portals and to write a positive Test Scenarios as well as Test cases for the above application

Code:

```
package kbexp;
public class kb {
   public static void main(String[] args) {

      String[] items = {
         "Home", "My Course", "Enrollment", "Attendance",
         "Assignment", "Examination", "Financial Record",
         "Disciplinary Record", "Raise Infra Issue", "My Profile"
      };

      String[] testCases = {
         "Check holiday notification",
         "View course status",
         "Select enrollment slot",
         "View attendance report",
         "View assignment details",
         "View examination details",
         "Check payment history",
         "View disciplinary details",
         "Submit infrastructure issue",
         "View profile details"
      };

      System.out.println("ARMS PORTAL - POSITIVE TEST CASES");
      System.out.println("-----------------------------------------------");

      System.out.printf("%-5s %-25s %-35s %-10s%n",
            "S.No", "Item", "Test Case", "Status");
```

```
        System.out.println("-----------------------------------------------");

        for (int i = 0; i < items.length; i++) {
            System.out.printf("%-5d %-25s %-35s %-10s%n",
                    i + 1, items[i], testCases[i], "PASS");
        }

        System.out.println("-----------------------------------------------");
        System.out.println("Test Approach: Manual Testing");
        System.out.println("Test Environment: Browser");
        System.out.println("Training Needed: No");
        System.out.println("Risk: Low");
        System.out.println("Test cases documented successfully.");
    }
}
```

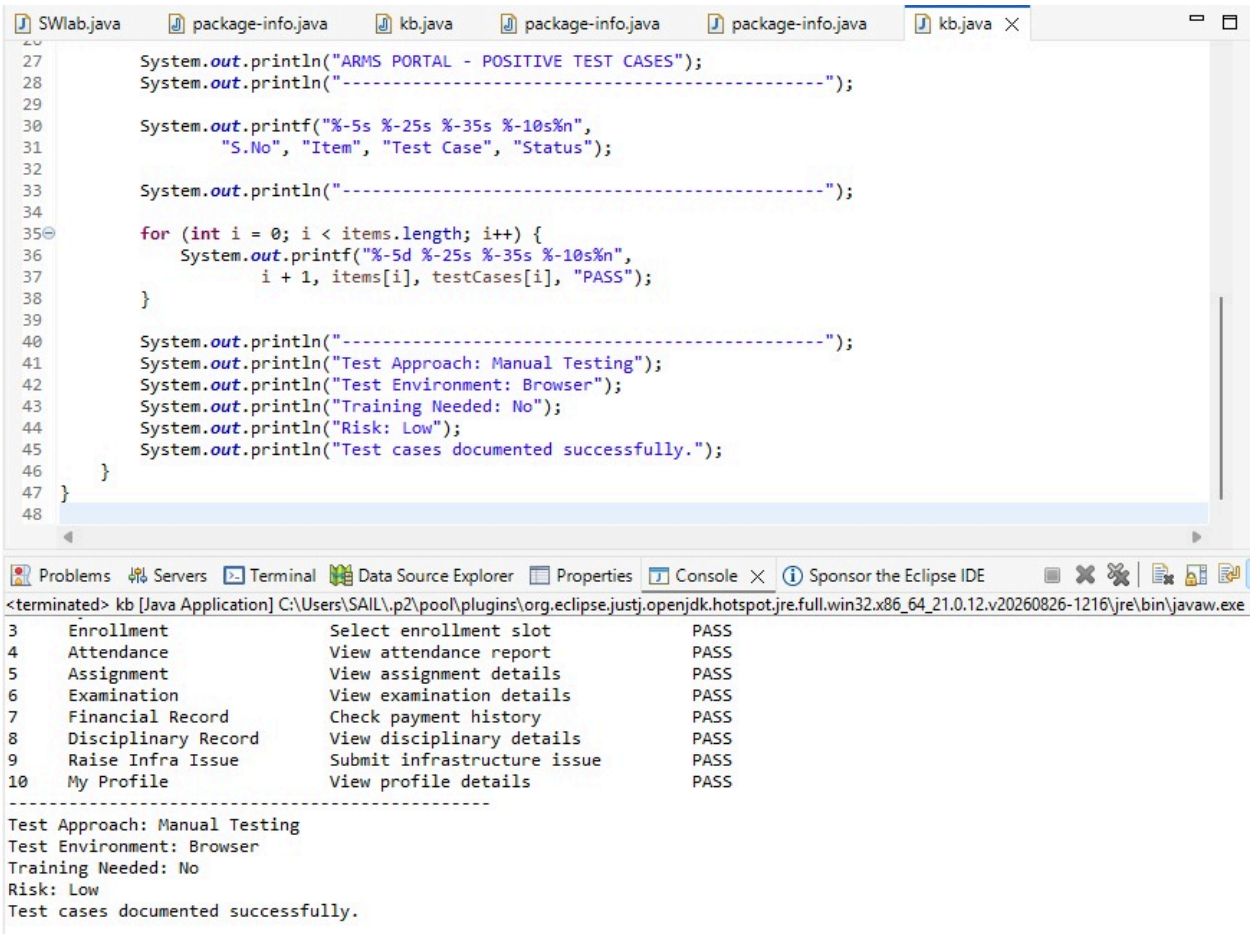
Report table:

S. No.	Item to be tested	Features	Approach	Test Data	Test Environment	Training Needed	Risk	Test Case / Scenario	Expected Output	Pass/Fail
1	Home	Holiday notification	Manual	No	Laptop/Browser	No	Low	Open ARMS and check the home page	Home page and notifications appear	Pass
2	My Course	Course status	Manual	No	Laptop/Browser	No	Low	Open My Course	Course status is displayed	Pass
3	Enrollment	Enrollment slot	Manual	Valid slot	Laptop/Browser	No	Low	Select an available slot	Slot is selected successfully	Pass
4	Attendance	Attendance report	Manual	Student details	Laptop/Browser	No	Low	Open attendance report	Attendance details appear	Pass
5	Assignment	Assignment details	Manual	Valid course	Laptop/Browser	No	Low	Open an assignment	Assignment details appear	Pass
6	Examination	Examination details	Manual	Student details	Laptop/Browser	No	Low	View examination details	Examination information appears	Pass
7	Financial Record	Payment history	Manual	Student account	Laptop/Browser	No	Low	Open payment history	Payment details appear	Pass
8	Disciplinary Record	Disciplinary details	Manual	Student account	Laptop/Browser	No	Low	Open disciplinary record	Record details appear, if available	Pass

9	Raise Infra Issue	Submit issue	Manual	Valid issue details	Laptop/Browser	No	Low	Submit an infrastructure issue	Issue is submitted successfully	Pass
10	My Profile	Profile details	Manual	Valid student account	Laptop/Browser	No	Low	Open My Profile	Profile details appear	Pass

Result: Positive test scenarios for the ARMS Portal were documented. Pass labels are illustrative and should be confirmed by testing the actual portal.

Code output:



3. To write a case study to identify & analyze the requirements of ARMS Portals and to write a negative Test Scenarios as well as Test cases for the above application.

Code:

```
package kbexp;
public class ARMSNegativeTest {
   public static void main(String[] args) {

      String[] items = {
         "Home",
         "My Course",
         "Enrollment",
         "Attendance",
         "Assignment",
         "Examination",
         "Financial Record",
         "Disciplinary Record",
         "Raise Infra Issue",
         "My Profile"
      };

      String[] testCases = {
         "Missing holiday information",
         "Invalid course details",
         "Invalid enrollment slot",
         "Unavailable attendance data",
         "Invalid assignment",
         "Expired examination link",
         "Invalid payment details",
         "Unauthorized record access",
         "Empty issue details",
         "Invalid profile information"
      };

      System.out.println("ARMS PORTAL - NEGATIVE TEST CASES");

      for (int i = 0; i < items.length; i++) {
         System.out.println("\nTest Case " + (i + 1));
         System.out.println("Item: " + items[i]);
         System.out.println("Scenario: " + testCases[i]);
```

```
            System.out.println("Expected Result: Error handled correctly");
        }

        System.out.println("\nNegative test cases documented successfully.");
    }
}
```

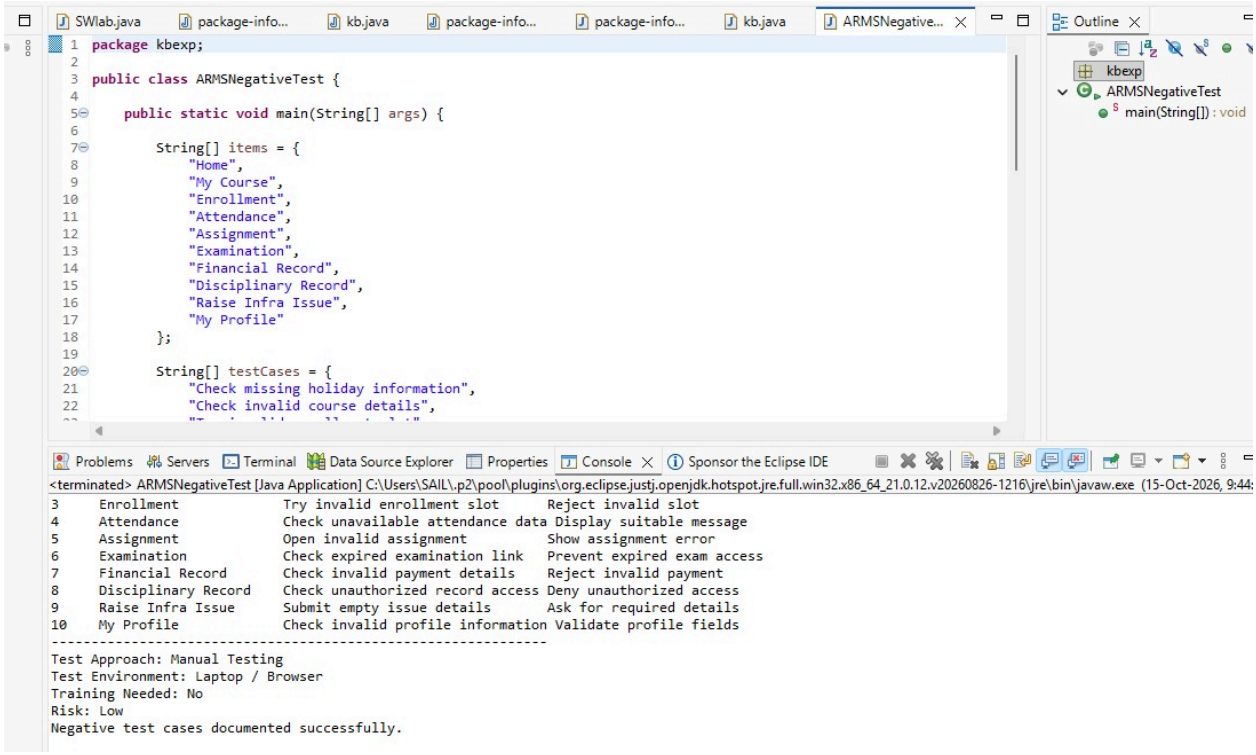
Report:

S.No.	Item to be tested	Features	Approach	Test Data	Test Environment	Training Needed	Risk	Test Case / Scenario	Expected Output	Pass/Fail
1	Home	Notifications	Manual	Missing data	Laptop/Browser	No	Low	Check home page when notifications are unavailable	Suitable message or empty state appears	To be tested
2	My Course	Course status	Manual	Invalid course details	Laptop/Browser	No	Low	Open invalid course details	Invalid course is rejected or an error appears	To be tested
3	Enrollment	Enrollment slot	Manual	Invalid slot	Laptop/Browser	No	Low	Select an invalid slot	Invalid slot is rejected	To be tested
4	Attendance	Attendance report	Manual	Unavailable data	Laptop/Browser	No	Low	Open unavailable attendance data	Suitable message appears	To be tested
5	Assignment	Assignment details	Manual	Invalid assignment	Laptop/Browser	No	Low	Open an invalid assignment	Error message appears	To be tested
6	Examination	Examination details	Manual	Expired link	Laptop/Browser	No	Low	Open an expired examination link	Access is prevented or an error appears	To be tested
7	Financial Record	Payment history	Manual	Invalid payment details	Laptop/Browser	No	Medium	Enter invalid payment details	Invalid payment is rejected	To be tested
8	Disciplinary Record	Record access	Manual	Unauthorized account	Laptop/Browser	No	Medium	Attempt unauthorized access	Access is denied	To be tested
9	Raise Infra Issue	Issue submission	Manual	Empty details	Laptop/Browser	No	Low	Submit an empty issue form	Required fields are indicated	To be tested

10	My Profile	Profile details	Manual	Invalid profile data	Laptop/Browser	No	Low	Save invalid profile information	Validation message appears	To be tested

Result: Negative test scenarios and expected outputs for the ARMS Portal were documented. Results are marked 'To be tested' because actual execution results were not supplied.

Code output:




```java
package kbexp;

public class ARMSNegativeTest {

    public static void main(String[] args) {

        String[] items = {
            "Home",
            "My Course",
            "Enrollment",
            "Attendance",
            "Assignment",
            "Examination",
            "Financial Record",
            "Disciplinary Record",
            "Raise Infra Issue",
            "My Profile"
        };

        String[] testCases = {
            "Check missing holiday information",
            "Check invalid course details",
```

4. To write a case study to identify & analyze the requirements of e-Commerce Applications and to write a positive Test Scenarios as well as Test cases for the above application.

Code:

```
package kbexp;
public class FlipkartNegativeTest {
   public static void main(String[] args) {

      String[] items = {
         "Home",
         "Search Button",
         "Shops",
         "Notifications",
         "Account",
         "Cart",
         "Account Settings",
         "My Cart",
         "Search Product",
         "Order Details"
      };

      String[] testCases = {
         "Open without login",
         "Search with empty input",
         "Open unavailable shop",
         "Check notifications without login",
         "Enter invalid account details",
         "Place order with empty cart",
         "Save profile with missing details",
         "Place order without address",
         "Search for unavailable product",
         "Cancel an invalid order"
      };

      String[] expectedResults = {
         "Ask user to log in if required",
         "Display validation or suitable results",
         "Display suitable message",
         "Ask user to log in if required",
```

```
            "Display validation message",
            "Prevent checkout with empty cart",
            "Validate required fields",
            "Ask user to add a valid address",
            "Display no products found",
            "Display suitable error message"
        };

        System.out.println("FLIPKART - NEGATIVE TEST CASES");
        System.out.println("-------------------------------------------------------------------------");

        System.out.printf("%-5s %-20s %-35s %-35s%n",
                "No.", "Item", "Negative Test Case", "Expected Result");

        System.out.println("-------------------------------------------------------------------------");

        for (int i = 0; i < items.length; i++) {
            System.out.printf("%-5d %-20s %-35s %-35s%n",
                    i + 1, items[i], testCases[i], expectedResults[i]);
        }

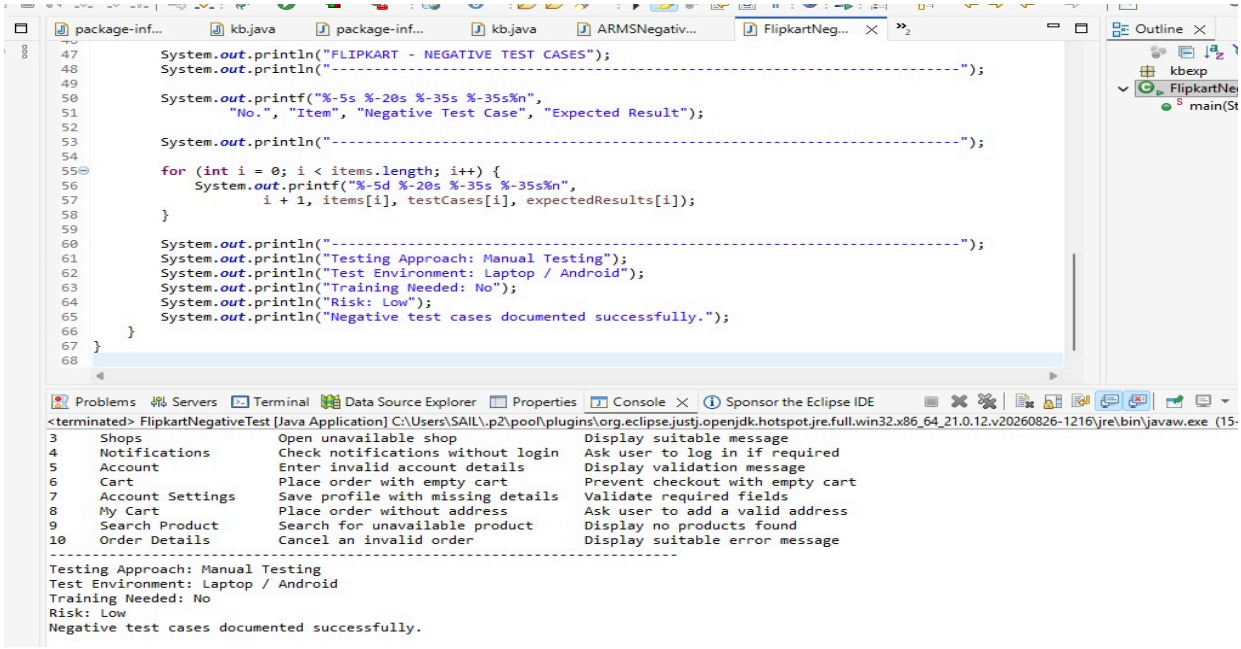
        System.out.println("-------------------------------------------------------------------------");
        System.out.println("Testing Approach: Manual Testing");
        System.out.println("Test Environment: Laptop / Android");
        System.out.println("Training Needed: No");
        System.out.println("Risk: Low");
        System.out.println("Negative test cases documented successfully.");
    }
}
```

Report:

S.No.	Item to be tested	Features	Approach	Test Data	Test Environment	Training Needed	Risk	Test Case / Scenario	Expected Output	Pass/Fail
1	Home	Email and password	Manual	Invalid login details	Laptop/Android	No	Low	Open Flipkart and enter incorrect login details	Incorrect email or password message appears	Pass
2	Search Button	Products, brands and shirts	Manual	Incorrect product	Laptop/Android	No	Low	Search for an incorrect product	Wrong or irrelevant products may appear	Pass
3	Shops	Mobiles and fashion	Manual	Unavailable mobile	Laptop/Android	No	Low	Search for mobiles in Shops	Searched mobiles do not appear	Pass
4	Trending Stores	Student and wedding stores	Manual	Store search	Laptop/Android	No	Low	Open a trending store	Another store may open	Pass
5	Notifications	Order notifications	Manual	Placed order	Laptop/Android	No	Low	Check notifications after placing an order	Order notification does not appear	Pass
6	Account	Orders, wishlist, coupons	Manual	Account details	Laptop/Android	No	Low	Open Account and Help Centre	Help Centre does not appear	Pass
7	Account	Logout	Manual	Logged-in account	Laptop/Android	No	Low	Click Logout	Flipkart does not log out	Pass
8	Account	Edit profile	Manual	Updated profile details	Laptop/Android	No	Low	Edit and save profile	Profile is not updated	Pass
9	Cart	Place order	Manual	Product in cart	Laptop/Android	No	Medium	Attempt to place an order	Order is not placed	Pass
10	Shops	Medicines	Manual	Medicine search	Laptop/Android	No	Low	Search for medicines	Medicines do not appear	Pass

Result: Negative test cases for the Flipkart e-commerce application were documented. The Pass labels reproduce the supplied lab-record statuses and are not independently verified.

Code output:


package-inf...
kb.java
package-inf...
kb.java
ARMSNegativ...
FlipkartNeg...
Outline
kbexp
FlipkartNe
main(St
System.out.println("FLIPKART - NEGATIVE TEST CASES");
System.out.println("----------");
System.out.printf("%-5s %-20s %-35s %-35s%n",
"No.", "Item", "Negative Test Case", "Expected Result");
System.out.println("----------");
for (int i = 0; i < items.length; i++) {
System.out.printf("%-5d %-20s %-35s %-35s%n",
i + 1, items[i], testCases[i], expectedResults[i]);
}
System.out.println("----------");
System.out.println("Testing Approach: Manual Testing");
System.out.println("Test Environment: Laptop / Android");
System.out.println("Training Needed: No");
System.out.println("Risk: Low");
System.out.println("Negative test cases documented successfully.");
}
}
Problems
Servers
Terminal
Data Source Explorer
Properties
Console
Sponsor the Eclipse IDE
<terminated> FlipkartNegativeTest [Java Application] C:\Users\SAIL\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.12.v20260826-1216\jre\bin\javaw.exe (15-
3 Shops Open unavailable shop Display suitable message
4 Notifications Check notifications without login Ask user to log in if required
5 Account Enter invalid account details Display validation message
6 Cart Place order with empty cart Prevent checkout with empty cart
7 Account Settings Save profile with missing details Validate required fields
8 My Cart Place order without address Ask user to add a valid address
9 Search Product Search for unavailable product Display no products found
10 Order Details Cancel an invalid order Display suitable error message
Testing Approach: Manual Testing
Test Environment: Laptop / Android
Training Needed: No
Risk: Low
Negative test cases documented successfully.

5. To write a case study to identify & analyze the requirements of e-Commerce Applications and to write a negative Test Scenarios as well as Tesi cases for the above application.

Code:

```
package kbexp;
public class FlipkartNegativeTest5 {
   public static void main(String[] args) {

      String[] items = {
         "Home",
         "Search Button",
         "Shops",
         "Trending Stores",
         "Notifications",
         "Account",
         "Logout",
         "Edit Profile",
         "Cart",
         "Medicines"
      };

      String[] testCases = {
         "Enter incorrect email or password",
         "Search for an incorrect product",
         "Search for unavailable mobiles",
         "Open a trending student store",
         "Check notifications for a placed order",
         "Open orders, wishlist and coupons",
         "Click the logout option",
         "Try updating the profile",
         "Try placing an order",
         "Search for medicines"
      };

      String[] expectedOutput = {
         "Incorrect email or password message",
         "Wrong products may appear",
         "Searched mobiles do not appear",
         "Another store may open",
```

```
            "Order notification does not appear",
            "Help Centre does not appear",
            "User is not logged out",
            "Profile is not updated",
            "Order is not placed",
            "Medicines do not appear"
        };

        String[] status = {
            "PASS", "PASS", "PASS", "PASS", "PASS",
            "PASS", "PASS", "PASS", "PASS", "PASS"
        };

        System.out.println("FLIPKART - NEGATIVE TEST CASES");

System.out.println("==========================================================
================");

        for (int i = 0; i < items.length; i++) {
            System.out.println("\nTest Case No: " + (i + 1));
            System.out.println("Item: " + items[i]);
            System.out.println("Test Scenario: " + testCases[i]);
            System.out.println("Expected Output: " + expectedOutput[i]);
            System.out.println("Status in Lab Manual: " + status[i]);
        }

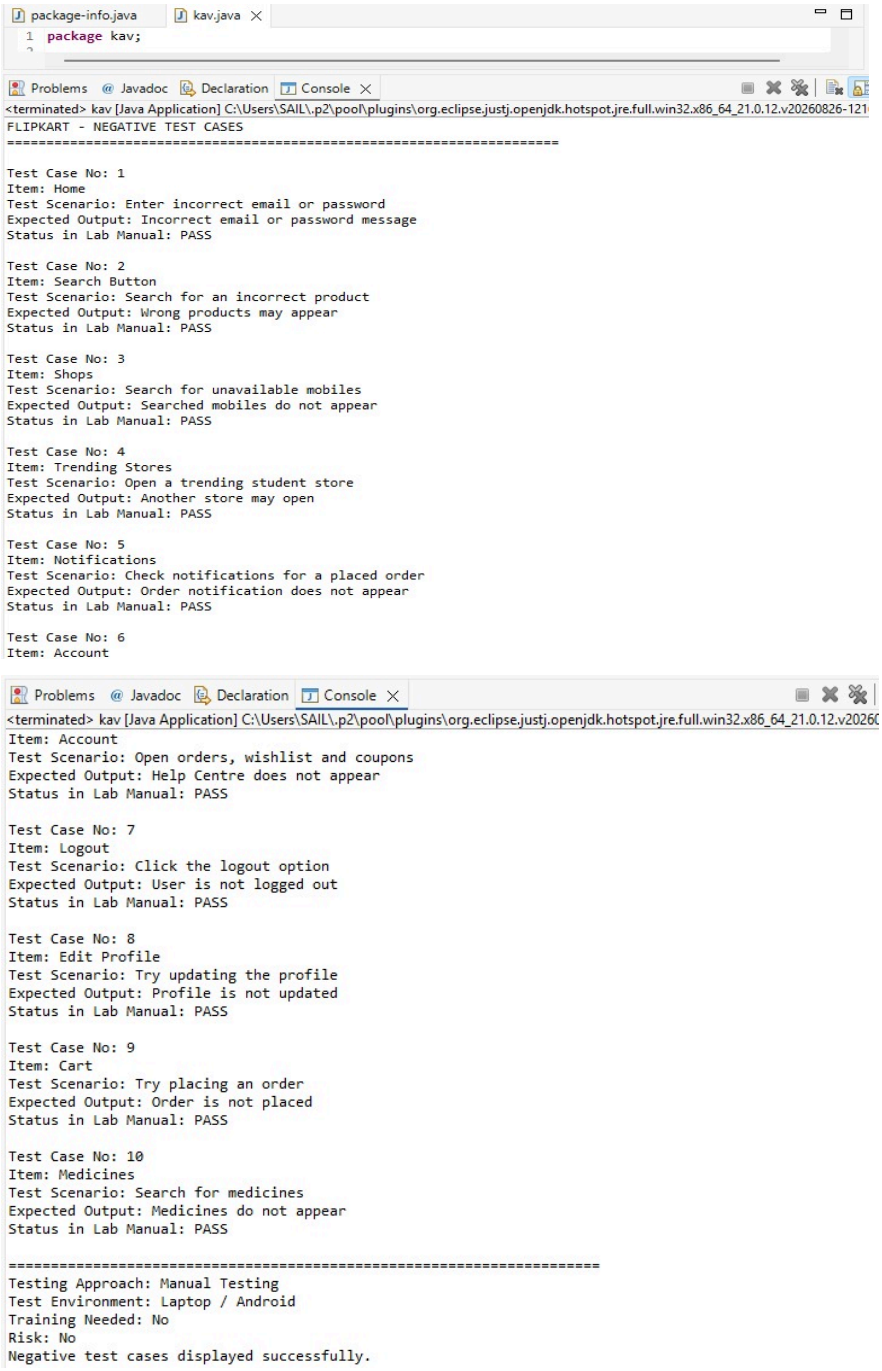

System.out.println("\n==========================================================
================");
        System.out.println("Testing Approach: Manual Testing");
        System.out.println("Test Environment: Laptop / Android");
        System.out.println("Training Needed: No");
        System.out.println("Risk: No");
        System.out.println("Negative test cases displayed successfully.");
    }
}
```

Report:

S. No.	Item to be tested	Features	Approach	Test Data	Test Environment	Training Needed	Risk	Test Case / Scenario	Expected Output	Pass/Fail
1	Home	Search, shops, notifications, account	Manual	Invalid email/password	Laptop/Android	No	Low	Open Flipkart and enter incorrect email and password	Incorrect email or password message appears	Pass
2	Search Button	Products, brands and shirts	Manual	Incorrect product	Laptop/Android	No	Low	Search for an incorrect product	Wrong products may appear	Pass
3	Shops	Mobiles, fashion, electronics	Manual	Mobile search	Laptop/Android	No	Low	Open Shops and search for mobiles	Searched mobiles do not appear	Pass
4	Trending Stores	Student and wedding stores	Manual	Store search	Laptop/Android	No	Low	Open Shops and select a trending store	Another store may open	Pass
5	Notifications	Order list	Manual	Placed order	Laptop/Android	No	Low	Check notifications for a placed order	Order notification does not appear	Pass
6	Account	Orders, wishlist, coupons	Manual	Account access	Laptop/Android	No	Low	Open Account and check Help Centre	Help Centre does not appear	Pass
7	Account	Logout	Manual	Logged-in account	Laptop/Android	No	Low	Open Account and click Logout	Flipkart does not log out	Pass
8	Account	Edit profile	Manual	Updated profile data	Laptop/Android	No	Low	Edit and save profile details	Profile is not updated	Pass
9	Cart	Place order	Manual	Product in cart	Laptop/Android	No	Low	Attempt to place an order	Order is not placed	Pass
10	Shops	Medicines	Manual	Medicine search	Laptop/Android	No	Low	Open Shops and search for medicines	Medicines do not appear	Pass

Result: Negative test cases for the Flipkart e-commerce application were documented. The Pass labels reproduce the supplied lab-record statuses and are not independently verified.

Code output:


package-info.java
kav.java
1 package kav;
Problems
Javadoc
Declaration
Console
<terminated> kav [Java Application] C:\Users\SAIL\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.12.v20260826-121
FLIPKART - NEGATIVE TEST CASES
Test Case No: 1
Item: Home
Test Scenario: Enter incorrect email or password
Expected Output: Incorrect email or password message
Status in Lab Manual: PASS
Test Case No: 2
Item: Search Button
Test Scenario: Search for an incorrect product
Expected Output: Wrong products may appear
Status in Lab Manual: PASS
Test Case No: 3
Item: Shops
Test Scenario: Search for unavailable mobiles
Expected Output: Searched mobiles do not appear
Status in Lab Manual: PASS
Test Case No: 4
Item: Trending Stores
Test Scenario: Open a trending student store
Expected Output: Another store may open
Status in Lab Manual: PASS
Test Case No: 5
Item: Notifications
Test Scenario: Check notifications for a placed order
Expected Output: Order notification does not appear
Status in Lab Manual: PASS
Test Case No: 6
Item: Account
Problems
Javadoc
Declaration
Console
<terminated> kav [Java Application] C:\Users\SAIL\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.12.v2026
Item: Account
Test Scenario: Open orders, wishlist and coupons
Expected Output: Help Centre does not appear
Status in Lab Manual: PASS
Test Case No: 7
Item: Logout
Test Scenario: Click the logout option
Expected Output: User is not logged out
Status in Lab Manual: PASS
Test Case No: 8
Item: Edit Profile
Test Scenario: Try updating the profile
Expected Output: Profile is not updated
Status in Lab Manual: PASS
Test Case No: 9
Item: Cart
Test Scenario: Try placing an order
Expected Output: Order is not placed
Status in Lab Manual: PASS
Test Case No: 10
Item: Medicines
Test Scenario: Search for medicines
Expected Output: Medicines do not appear
Status in Lab Manual: PASS
Testing Approach: Manual Testing
Test Environment: Laptop / Android
Training Needed: No
Risk: No
Negative test cases displayed successfully.